

Case Study: PCA for Retail Market Entry Strategy

Undergraduate Statistics Program | Principal Component Analysis in R

Background

- A national retail company wants to identify European cities that are most attractive for expansion using multiple market indicators.
- A strategy and analytics team has assembled a city-level dataset with economic, demographic, consumer, and infrastructure indicators for a set of European cities. Examples of variables include income, population growth, tourism, digital spending, public transport, rent levels, and competitor density.
- Because many of these indicators are strongly correlated, a simple ranking based on the raw variables may be misleading. The client therefore asks for a Principal Component Analysis (PCA) that reduces the dimensionality of the problem, identifies the main market dimensions, and supports a defensible recommendation on expansion priorities.

Assignment objective

Your task is to analyse the dataset in R and produce a short report for the client. You should use PCA to study the relationships among the evaluation criteria, interpret the retained components, compare cities, and discuss how the results could be used to support a retail market-entry strategy.

By the end of the assignment, students should be able to:

- compute and comment on multivariate descriptive statistics;
- perform PCA in R on centred and standardised data;
- interpret loadings, explained variance, and component scores;
- group variables and cities using PCA results;
- translate statistical findings into a business recommendation.

Task 1. Compute useful descriptive measures for the multivariate dataset.

Present summary statistics and visualisations that help describe the city-level data. Useful outputs may include means, standard deviations, minima and maxima, a correlation matrix, and suitable plots. Briefly discuss which market indicators show high variability, which ones appear strongly correlated, and whether there is evidence of redundant information.

Task 2. Use Principal Component Analysis for standardized data to study the relationships among the variables.

Report the coefficients (loadings) of all principal components and the percentage of total variance explained by each component. Include a scree plot and a concise interpretation of the overall component structure.

Task 3. Decide how many principal components should be retained.

Justify the number of retained components. Use and compare multiple criteria. What business interpretation can be given to the first two principal components? Compute the scores of these components for all cities and comment on the cities with the highest and lowest scores.

Task 4. How can the retail evaluation criteria be grouped based on the PCA results?

How good is the grouping?

Use the loadings and/or variable plots to identify groups of variables that express similar dimensions of market attractiveness. Comment on whether the grouping is clear and whether some variables contribute ambiguous or mixed information.

Task 5. How can the cities be grouped based on these results?

Use the component scores to explore whether cities form meaningful groups. Are there visible differences between Northern and Southern European cities? Discuss any regional patterns carefully and avoid overclaiming when separation is weak.

Task 6. How could the PCA results be used to rank the cities in descending order of market attractiveness?

Propose a defensible composite ranking method based on one or more retained principal components. Explain the logic of your ranking rule, present the cities that appear most and least attractive for expansion, and discuss advantages and limitations of using PCA for ranking.

Task 7. How much do the results change when standardised data are used?

Repeat the PCA using centred-only data and compare the results with the analysis based on standardised data. Comment on changes in loadings, explained variance, interpretation, city grouping, and the final ranking. Explain when standardisation is preferable in this case.

Required deliverables

No.	Deliverable	Description
1	Short report (5–8 pages)	Business context, descriptive analysis, PCA results, interpretation, grouping of variables and cities, ranking proposal, and comparison between centred and standardised analyses.
2	R script or R Markdown / Quarto file	Fully reproducible code, clearly commented, with clean data preparation and labelled outputs.
3	Executive summary (7-8 ppt slides)	A concise recommendation to the client on which types of cities appear most attractive for retail expansion and why.

Recommended workflow

- Inspect the variables, identify non-numeric columns, and check missing values.
- Decide whether imputation, row filtering, or variable removal is necessary.
- Run PCA first on standardised data, then on centred data.
- Use loadings and scores to build a narrative for market attractiveness.
- End with a recommendation aimed at a non-technical consulting client.

Assessment criteria

- Correctness of the statistical analysis.
- Quality of interpretation and business reasoning.
- Clarity of tables, plots, and written discussion.
- Reproducibility and readability of the R code.

Interview relevance

- This case mirrors the kind of work often discussed in consulting, analytics, and strategy interviews: handling correlated indicators, simplifying a complex decision problem, and translating technical outputs into an expansion recommendation for management.